BTC-004 Independent Referee Audit

Dossier: *Morse-Index Rigidity for Even Ternary Quartics*
Review status: AI-assisted mathematical review; not external human peer review.
Novelty: not established.

Overall verdict

REVISE. The proposition is plausible and the local algebra is largely correct, but the proof as submitted does not yet justify its exhaustive case split. The main missing step is the assertion that, when the reduced interior Hessian K is indefinite, the three edge curvatures cannot all have the same sign. That assertion is not a consequence of indefiniteness of an arbitrary two-dimensional quadratic form alone. It can be repaired using the global Morse relation on $\mathbb{RP}^2$, but that repair should be written explicitly.

I found no counterexample to the stated two index triples. The corrected argument supports the proposition under the stated hypotheses.

1. Support count: essentially PASS

For

$$f(x,y,z) = d_1x^4 + d_2y^4 + d_3z^4 + ax^2y^2 + bx^2z^2 + cy^2z^2,$$

on the unit sphere set $u = x^2, v = y^2, w = z^2$, so $u, v, w \geq 0$ and $u + v + w = 1$. Then

$$Q(u,v,w) = d_1u^2 + d_2v^2 + d_3w^2 + auv + buw + cvw.$$

A point with exactly k nonzero coordinates lifts from a positive squared-coordinate tuple to 2^{k-1} projective points. The three simplex vertices are always critical. Each open edge has at most one stationary point unless its restriction is constant, and such a point contributes two projective critical points. If the affine interior Hessian is nonsingular, there is at most one interior stationary tuple, contributing four projective points.

Under the Morse hypothesis, constant edge restrictions and degenerate critical points are excluded. Therefore 13 projective critical points force exactly

$$3 + 3 \cdot 2 + 4 = 13,$$

namely all three vertices, one stationary point on each open edge, and one full-support stationary point with four sign lifts. This part is correct, provided “Morse” is understood

for the normalized function on $\mathbb{RP}^2$ and all boundary degeneracies are explicitly excluded.

2. Edge-normal identity: PASS

For the xy-edge use $t = u, n = w, v = 1 - t - n$. The constant Hessian is

$$K = \begin{pmatrix} \kappa & h \\ h & \ell \end{pmatrix}, \quad \kappa = 2(d_1 + d_2 - a), \quad h = 2d_2 - a + b - c, \quad \ell = 2(d_2 + d_3 - c).$$

If (t_*, n_*) is the interior stationary point and $(t_e, 0)$ is the edge stationary point, then

$$\kappa(t_e - t_*) - h n_* = 0.$$

Substitution into the inward derivative gives

$$Q_n(t_e, 0) = -\frac{\det K}{\kappa} n_*.$$

Since $n_* > 0$ and $\kappa \neq 0$,

$$\operatorname{sgn} Q_n(t_e, 0) = -\operatorname{sgn}(\det K)\operatorname{sgn}(\kappa).$$

The derivation is valid for arbitrary coefficients and applies cyclically to the other edges.

3. Index transfer: PASS with convention clarification

At an open-edge critical point, the tangent direction along the edge has Hessian sign $\operatorname{sgn} \kappa$. The missing projective coordinate enters quadratically through the squared-coordinate variable, so the transverse Hessian sign is $\operatorname{sgn} Q_n$. The factor relating the squared-coordinate derivative to the projective Hessian is positive and does not change signs.

At a vertex, if an adjacent edge has an interior stationary point, the inward endpoint derivative has sign $-\operatorname{sgn} \kappa$. Thus the corresponding projective Hessian direction has that same sign. If the Morse index is the number of negative Hessian eigenvalues, the vertex index equals the number of incident edges with positive curvature. The dossier should state this convention explicitly because its prose alternates between “maxima/minima” and index counts.

4. The proof gap in the indefinite case

When $\det K > 0$, K is definite. The dossier’s conclusions follow:

- positive-definite K: four interior minima, six edge saddles, three vertex maxima, giving $(m_0, m_1, m_2) = (4, 6, 3)$;
- negative-definite K: four interior maxima, six edge saddles, three vertex minima, giving $(3, 6, 4)$.

When $\det K < 0$, K is indefinite. The edge identity implies that an edge with $\kappa > 0$ is a local minimum and an edge with $\kappa < 0$ is a local maximum. The interior points are saddles.

The submitted proof then says that the three edge curvatures cannot all be positive or all negative, referring vaguely to an Euler relation. This needs correction. Indefiniteness of a two-dimensional quadratic form alone does **not** rule out a prescribed quadratic form having the same sign on three arbitrary directions. The required exclusion follows from the global Morse relation after the corresponding index counts are computed.

If all three κ's were positive, the six edge points would be minima, the three vertices maxima, and the four full-support points saddles. Hence

$$m_0 - m_1 + m_2 = 6 - 4 + 3 = 5,$$

contradicting

$$m_0 - m_1 + m_2 = \chi(\mathbb{RP}^2) = 1.$$

If all three κ's were negative, the roles of minima and maxima reverse and the Euler characteristic is again violated. Since no edge curvature is zero under the Morse hypothesis, the only remaining patterns are one positive/two negative or two positive/one negative.

Those two patterns give respectively

$$(3, 6, 4) \quad \text{or} \quad (4, 6, 3).$$

This repairs the exhaustive classification.

5. Boundary and degeneracy cases

The dossier is correct that an edge curvature $\kappa = 0$ makes the edge restriction constant or otherwise destroys the claimed isolated Morse structure. If an edge stationary point reaches a vertex, or a full-support stationary point reaches an edge, the local decomposition changes and the relevant critical point is degenerate in the projective coordinates. These cases are incompatible with the 13-distinct-point Morse hypothesis, but the proof should explicitly state why the relevant Hessian loses rank.

The proof should also say that the affine interior Hessian K is nonsingular because a full-support stationary point exists and is Morse. If K were singular, the quadratic gradient system would either have no isolated interior solution or a non-isolated solution set.

6. Attainment

The example

$$f = 2x^4 + y^4 + z^4$$

is in the even-power family and has the 13-point support count. Its index triple is consistent with $(4, 6, 3)$ under the convention $m_i = \#$ points of Morse index i, where index is the number of negative Hessian directions. Replacing f by $-f$ exchanges minima and maxima and realizes $(3, 6, 4)$.

7. Prior-art comparison

Maccioni' s paper proves lower bounds and realizability results for the number of real eigenvectors of real ternary forms. Its Proposition 5.2 states that ternary quartics can attain 13 real eigenvectors and gives examples, including the Fermat quartic. Its Morse discussion uses the Euler relation on the sphere to constrain total critical-point counts. I did not find in the checked text a classification of Morse-index triples for the restricted even-in-each-coordinate family.

Cartwright–Sturmfels and Abo–Seigal–Sturmfels establish the algebraic eigenpoint count and eigenconfiguration framework, including the generic count 13 for ternary quartics. They do not, in the checked portions, give this even-power Morse-index classification.

Therefore the proposition' s exact index-rigidity statement appears **not directly subsumed by the checked references**, but this is not an exhaustive novelty clearance. The result is a specialized elementary classification built from Morse theory and quadratic-simplex geometry; any novelty claim should remain provisional until a broader literature search, including real algebraic geometry and equivariant Morse theory, is completed.

Disposition

- Support count: **PASS**.
- Edge-normal identity: **PASS**.
- Index transfer: **PASS with convention clarification**.
- Exhaustive indefinite-case split: **REVISE; Euler-characteristic repair required**.

- Degenerate/boundary treatment: **PASS after explicit rank-loss wording**.
- Attainment: **PASS**.
- Prior art: **no exact equivalent found in the checked sources; not a novelty clearance**.

Final recommendation: revise the proof by replacing the current indefinite-sign paragraph with the explicit Morse/Euler contradiction above, state the index convention, and then seek a fresh independent review. Do not call this an established novel theorem yet.

References

[1] Mauro Maccioni, The number of real eigenvectors of a real polynomial
[2] H. Abo, A. Seigal, and B. Sturmfels, Eigenconfigurations of Tensors